

Exercises

Linear models - Model selection

In the previous exercise you fitted a pre-conceived model which included the main effects of the area of the forest patch (**LOGAREA**), the grazing intensity (**FGRAZE**) and the interaction between these two explanatory variables (**FGRAZE:LOGAREA**). This was useful as a training exercise, and might be a viable approach when analysing these data if an experiment had been designed to test these effects only. However, if other potentially important variables are not included in the model this may lead to biased inferences (interpretation). Additionally, if the goal of the analysis is to explore what models explain the data in a parsimonious way (as opposed to formally testing hypotheses), we would also want to include relevant additional explanatory variables.

Here we revisit the previous loyn data analysis, and ask if a ‘better’ model for these data could be achieved by including additional explanatory variables and by performing model selection. Because we would like to test the significance of the interaction between **LOGAREA**, and **FGRAZE**, whilst accounting for the potential effects of other explanatory variables, we will also include **LOGAREA**, **FGRAZE** and their interaction (**FGRAZE:LOGAREA**) in the model as before. Including other interaction terms between other variables may be reasonable, but we will focus only on the **FGRAZE:LOGAREA** interaction as we have relatively little information in this data set (67 observations). This will hopefully avoid fitting an overly complex model which will estimate many parameters for which we have very little data. This is a balance you will all have to maintain with your own data and analyses (or better still, perform a power analysis before you even collect your data). No 4-way interaction terms in your models please!

It’s also important to note that we will assume that all the explanatory variables were collected by the researchers because *they believed* them to be biologically relevant for explaining bird abundance (i.e. data were collected for a reason). Of course, this is probably not your area of expertise but it is nevertheless a good idea to pause and think what might be relevant or not-so relevant and why. This highlights the importance of knowing your study organism / study area and discussing research designs with colleagues and other experts in the field before you collect your data. What you should try to avoid is collecting heaps of data across many variables (just because you can) and then expecting your statistical models to make sense of it for you. As mentioned in the lecture, model selection is a relatively controversial topic and should not be treated as a purely mechanical process (chuck everything in and see what comes out). Your expertise needs to be woven into this process otherwise you may end up with a model that is implausible or not very useful (and all models need to be useful!).

1. Import the ‘loyn.txt’ data file into RStudio and assign it to a variable called **loyn**. Here we will be using all the explanatory variables to explain the variation in bird density. If needed, remind yourself of your data exploration you conducted previously. Do any of the remaining variables need transforming (i.e. **AREA**, **DIST**, **LDIST**) or converting to a factor type variable (i.e. **GRAZE**)? Add the transformed variables to the **loyn** dataframe.

2. Let's start with a very quick graphical exploration of any potential relationships between each explanatory variable (collinearity) and also between our response and explanatory variables (what we're interested in). Create a pairs plot using the function `pairs()` of your variables of interest. Hint: restrict the plot to the variables you actually need. An effective way of doing this is to store the names of the variables of interest in a vector `VOI <- c("Var1", "Var2", ...)` and then use the naming method for subsetting the data set `Mydata[, VOI]`. If you feel like it, you can also add the correlations to the lower triangle of the plot as you did previously (don't forget to define the function first).

3. Now, let's fit our maximal model. Start with a model of `ABUND` and include all explanatory variables as main effects. Also include the interaction `LOGAREA:FGRAZE` but no other interaction terms as justified in the preamble above. Don't forget to include the transformed versions of the variables where appropriate (but not the untransformed variables as well otherwise you will have very strong collinearity between these variables!). Perhaps, call this model `M1`.

4. Have a look at the summary table of the model using the `summary()` function. You'll probably find this summary is quite complicated with lots of parameter estimates (14) and P values testing lots of hypotheses. Are all the P values less than our cut-off of 0.05? If not, then this suggests that some form of model selection is warranted to simplify our model.

Before you go any further though, a word of caution about looking at 14 tests all at once. If none of these explanatory variables had any real effect on bird abundance whatsoever, roughly how many of the 14 P values would you still expect to fall below 0.05 purely by chance?

5. Let's perform a first step in model selection using the `drop1()` function and use an F test based model selection approach. This will allow us to decide which explanatory variables may be suitable for removal from the model. Remember to use the `test = "F"` argument to perform F tests when using `drop1()`. Which explanatory variable is the best candidate for removal and why?

What hypothesis is being tested when we do this model selection step?

A note on keeping track of model selection. Describing what you did during model selection is genuinely difficult to do well in writing, and it is done badly in the published literature more often than almost anything else on this course. Papers routinely say "a minimum adequate model was obtained by backward selection" and leave the reader with no idea what was in the starting model, what came out, in what order, or on what grounds.

A summary table solves this, and it is worth seeing one being built rather than just being shown a finished one. So as we work through the deletions below I'll add a row to a table at each step, and you'll find it updated in the solution to each question. By the time we reach the end we'll have something that could go straight into a results section. You don't need to construct it yourself here, just follow along, but do consider making one like it when you report a model selection of your own.

Here is the structure, with the starting model in it:

Table 1: Summary of model selection.

Model	Terms in the model	Term removed	df	F	P
M1	LOGLDIST + LOGDIST + YR.ISOL + ALT + LOGAREA + FGRAZE + LOGAREA:FGRAZE	–	–	–	–

Two things about that first row. The starting model is in the table because a reader cannot judge a selection procedure without knowing what it started from: dropping four terms from a model of six is a very different exercise from dropping four from a model of twenty. And the F and P columns hold dashes because nothing was removed in order to arrive at M1. Those columns describe the deletion that got you from the previous model to this one, and for the first row there isn't one.

6. Update and refit your model and remove the least significant explanatory variable (from above). Repeat single term deletions with `drop1()` again using this updated model. You can update the model by just fitting a new model without the appropriate explanatory variable and assign it to a new name (M2). Alternatively you can use the `update()` function instead (I show you how to do this in the solutions).

7. Again, update the model to remove the least significant explanatory variable (from above) and repeat single term deletions with `drop1()`.

8. Once again, update the model to remove the least significant explanatory variable (from above) and repeat single term deletions with `drop1()`.

9. And finally, update the model to remove the least significant explanatory variable (from above) and repeat single term deletions with `drop1()`.

10. If all goes well, your final model should be `lm(ABUND ~ LOGAREA + FGRAZE + LOGAREA:FGRAZE)` which you encountered in the previous exercise. Also, you may have noticed that the output from the `drop1()` function does not include the main effects of `LOGAREA` or `FGRAZE`. Can you think why this might be the case?

11. Now that you have your final model, you should go through your model validation and model interpretation as usual. As we have already completed this in the previous exercise I'll leave it up to you to decide whether you include it here (you should be able to just copy and paste the code).

Please make sure you understand the biological interpretation of each of the parameter estimates and the interpretation of the hypotheses you are testing. As before, lead with the estimate and its confidence interval (`confint()`) and treat the P value as supporting evidence rather than as the finding (reminder).

There is also one caveat that applies to your final model here but did not apply to the model you fitted in the previous exercise, even though the two models are identical. Have a think about it before you look at the solutions. The P values and confidence intervals for your final model are calculated as though you had specified that model in advance. You didn't: you arrived at it by looking at these same data. Does that make them too optimistic, too conservative, or does it make no difference at all?

One last caution, and it applies to everything you have done in this exercise. These are observational data. Nobody assigned patch areas or grazing intensities to these forests, they were measured as they were found. A linear model will happily estimate an effect of grazing on bird abundance, but it cannot tell you that grazing *causes* that change.

Two things are worth keeping in mind. First, variables you didn't measure may be doing the work. Heavily grazed patches might also be the ones nearest roads, or on more fertile soil, or with a different understorey, and any of those could be the real driver of the difference in bird abundance.

Second, your explanatory variables are correlated with one another, as the pairs plot in Question 2 showed. That has a specific consequence for how you read the output: every estimate in your model is the effect of that variable *holding the other variables in the model constant*. Change which variables are in the model and the estimate changes, as you saw during model selection. So a coefficient is never simply “the effect of grazing”; it is the effect of grazing after accounting for whatever else you happened to include, and it will absorb some of the effect of any correlated variable you left out.

None of this means observational studies aren’t worth doing, and most of ecology is observational for perfectly good reasons. It does mean your language should match the evidence you actually have. “Bird abundance was lower in more heavily grazed patches” is supported by these data. “Grazing reduces bird abundance” is a stronger claim than a model like this one can carry on its own, and it needs experimental evidence, or a well understood mechanism, standing behind it.

12. Now for reporting your results, and this is the last write-up of the course. You arrived at your final model by a selection procedure, which brings a couple of reporting problems the previous three exercises didn’t have.

This is where the table you have been building since Question 5 earns its keep. Present it as a table in your results, exactly as you would in a paper, and then **refer to it** in your text rather than describing every deletion in prose. That is what the table is for: it carries the detail, so your paragraph can carry the argument.

Write **no more than 200 words** of accompanying text for the methods and results of a paper. Your reader is an ecologist who knows their stuff but isn’t a statistician, and who wasn’t in the room while you did the analysis. They need to be able to work out from your text and your table exactly what you did and how much to trust it. Questions 4 and 11 have already told you the two things an honest account has to include that a naive one would leave out.

OPTIONAL questions if you have time / energy / inclination!

A1. If we weren’t aiming to directly test the effect of the `LOGAREA:FGRAZE` interaction statistically (i.e. test this specific hypothesis), we could use AIC to perform model selection. Repeat the model selection you did above, but this time use the `drop1()` function and perform model selection using AIC instead. Don’t forget, if we want to perform model selection based on AIC with the `drop1()` function we need to omit the `test = "F"` argument)

A table again, and two ways of reading AIC. As with the F test route, I’ll build a summary table as we go. Table 2 has the same shape as Table 1, but with AIC in place of the F statistic and P value, and with two AIC columns rather than one. Those two columns carry the same evidence read in opposite directions, so it is worth being clear about them before you see any numbers.

AIC change is what happened to AIC when a term was deleted. This is the number the selection decision is actually based on, and it is what `drop1()` puts in front of you. A negative value means AIC fell when the term came out, so the simpler model was preferred.

Δ AIC is the column you will see in published tables: each model’s AIC minus the **lowest** AIC in the set. The preferred model therefore has Δ AIC = 0, and every other model has a positive value.

Notice what that implies about *when* you can build the column. You cannot work out Δ AIC until you know which model has the lowest AIC, and you don’t know that until selection has finished. In practice, then, you would construct this table at the end and present it once. I am showing it growing row by row purely so you can see how it is assembled, and the Δ AIC values below are all calculated against the final best model, which is why none of the intermediate tables has a zero in that column.

One more thing about presentation. I have put both AIC columns in here so that you can see the two readings side by side, but you would **not** normally do that in a thesis or a paper. Pick one, and it is almost always ΔAIC , because that is what readers expect and it does not depend on the order in which you happened to delete things.

The usual rule of thumb (Burnham and Anderson 2002) applies to both columns, but reads differently in each:

- **AIC change:** a change of less than 2 either way is no real support for preferring either model, so the simpler one wins and the term goes. A deletion that *raises* AIC by roughly 4 to 7 is considerable support for keeping the term, and by more than 10, strong support.
- **ΔAIC :** a model within 2 of the best has substantial support, one 4 to 7 above the best has considerably less, and one more than 10 above has essentially none.

k , in the third column, is simply the number of parameters the model estimates.

A2. Refit your model with the variable associated with the lowest AIC value removed. Run `drop1()` again on your updated model. Perhaps call this new model `M2.AIC`.

A3. Refit your model with the variable associated with the lowest AIC value removed and run `drop1()` again on your new model (`M3.AIC`).

A4. Repeat your model selection by removing the variable indicated by the model with the lowest AIC.

A5. Rinse and repeat as above.

If all goes well, your final model should be `lm(ABUND ~ LOGAREA + FGRAZE + LOGAREA:FGRAZE)`. This is the same model you ended up with when using the F test based model selection. This might not always be the case and generally speaking AIC based model selection approaches tend to favour more complicated minimum adequate models compared to F test based approaches.

We don't need to re-validate or re-interpret the model, since we have already done this previously.

You have already seen how to present this. Table 2 above is the AIC equivalent of Table 1, and either one can go straight into a results section with the text referring to it, exactly as in Question 12.

A couple of practical notes if you build one of these for your own work. The AIC values reported by `drop1()` are not the same as those from the `AIC()` function, because the two calculate AIC on slightly different scales. Neither is wrong, but never mix values from the two in the same table. And if you want a formatted table in a report rather than console output, `knitr::kable()` will do it (see the 'Installing R Markdown' appendix of the Introduction to R book).

References

Burnham, K.P. and Anderson, D.R. (2002) *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*, 2nd edition. Springer, New York.

End of the model selection exercise